

STAT405 Bayesian Statistics Final Project

EDA

```
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(corrplot)

## corrplot 0.95 loaded

library(car)

## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##   recode

house_price <- read.csv("~/Downloads/STAT_405/house-prices-advanced-regression-techniques/train.csv")

# check MSSubClass, it should be character
house_price$MSSubClass <- as.character(house_price$MSSubClass)

# Replace NA with explicit categories
house_price$Alley[is.na(house_price$Alley)] <- "NoAlley"
house_price$GarageType[is.na(house_price$GarageType)] <- "NoGarageType"
house_price$MasVnrType[is.na(house_price$MasVnrType)] <- "NoMasVnrType"
house_price$MasVnrArea[is.na(house_price$MasVnrArea)] <- 0
house_price$Fence[is.na(house_price$Fence)] <- "NoFence"
house_price$MiscFeature[is.na(house_price$MiscFeature)] <- "NoMiscFeature"

house_price$BsmtQual[is.na(house_price$BsmtQual)] <- "NoBsmt"
house_price$BsmtCond[is.na(house_price$BsmtCond)] <- "NoBsmt"
house_price$BsmtExposure[is.na(house_price$BsmtExposure)] <- "NoBsmt"
house_price$BsmtFinType1[is.na(house_price$BsmtFinType1)] <- "NoBsmt"
house_price$BsmtFinType2[is.na(house_price$BsmtFinType2)] <- "NoBsmt"
```

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house_price$FireplaceQu[is.na(house_price$FireplaceQu)] <- "NoFireplace"
house_price$GarageFinish[is.na(house_price$GarageFinish)] <- "NoGarage"
house_price$GarageQual[is.na(house_price$GarageQual)] <- "NoGarage"
house_price$GarageCond[is.na(house_price$GarageCond)] <- "NoGarage"
house_price$PoolQC[is.na(house_price$PoolQC)] <- "NoPool"

# Add indicator variables first
house_price$HasBsmt <- as.factor(ifelse(house_price$TotalBsmtSF > 0, 1, 0))
house_price$HasGarage <- as.factor(ifelse(house_price$GarageArea > 0, 1, 0))
house_price$HasFireplace <- as.factor(ifelse(house_price$Fireplaces > 0, 1, 0))
house_price$HasPool <- as.factor(ifelse(house_price$PoolArea > 0, 1, 0))
house_price$HasMasVnr <- as.factor(ifelse(house_price$MasVnrArea > 0, 1, 0))

# Convert to ordered factors
qual_levels <- c("Po", "Fa", "TA", "Gd", "Ex")
house_price$ExterQual <- factor(house_price$ExterQual,
                                levels = qual_levels, ordered = TRUE)
house_price$ExterCond <- factor(house_price$ExterCond,
                                levels = qual_levels, ordered = TRUE)
house_price$BsmtQual <- factor(house_price$BsmtQual,
                                levels = c("NoBsmt", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)
house_price$BsmtCond <- factor(house_price$BsmtCond,
                                levels = c("NoBsmt", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)
house_price$BsmtExposure <- factor(house_price$BsmtExposure,
                                    levels = c("NoBsmt", "No", "Mn", "Av", "Gd"),
                                    ordered = TRUE)
house_price$BsmtFinType1 <- factor(house_price$BsmtFinType1,
                                    levels = c("NoBsmt", "Unf", "LwQ", "Rec", "BLQ", "ALQ", "GLQ"),
                                    ordered = TRUE)
house_price$BsmtFinType2 <- factor(house_price$BsmtFinType2,
                                    levels = c("NoBsmt", "Unf", "LwQ", "Rec", "BLQ", "ALQ", "GLQ"),
                                    ordered = TRUE)
house_price$HeatingQC <- factor(house_price$HeatingQC,
                                levels = qual_levels, ordered = TRUE)
house_price$KitchenQual <- factor(house_price$KitchenQual,
                                levels = qual_levels, ordered = TRUE)
house_price$Functional <- factor(house_price$Functional,
                                levels = c("Sal", "Sev", "Maj2", "Maj1", "Mod", "Min2", "Min1", "Typ"),
                                ordered = TRUE)
house_price$FireplaceQu <- factor(house_price$FireplaceQu,
                                levels = c("NoFireplace", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)
house_price$GarageFinish <- factor(house_price$GarageFinish,
                                levels = c("NoGarage", "Unf", "RFn", "Fin"),
                                ordered = TRUE)
house_price$GarageQual <- factor(house_price$GarageQual,
                                levels = c("NoGarage", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)
house_price$GarageCond <- factor(house_price$GarageCond,
                                levels = c("NoGarage", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)

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house_price$PoolQC <- factor(house_price$PoolQC,
                             levels = c("NoPool", "Fa", "TA", "Gd", "Ex"),
                             ordered = TRUE)
house_price$LotShape <- factor(house_price$LotShape,
                               levels = c("IR3", "IR2", "IR1", "Reg"),
                               ordered = TRUE)
house_price$LandSlope <- factor(house_price$LandSlope,
                                 levels = c("Sev", "Mod", "Gtl"),
                                 ordered = TRUE)
house_price$PavedDrive <- factor(house_price$PavedDrive,
                                  levels = c("N", "P", "Y"),
                                  ordered = TRUE)

# Convert ordered factors to numeric scores
ordinal_to_numeric <- function(x) as.numeric(x) - 1

ord_vars <- c(
  "ExterQual", "ExterCond",
  "BsmtQual", "BsmtCond", "BsmtExposure", "BsmtFinType1", "BsmtFinType2",
  "HeatingQC", "KitchenQual", "Functional", "FireplaceQu",
  "GarageFinish", "GarageQual", "GarageCond",
  "PoolQC", "LotShape", "LandSlope", "PavedDrive"
)
house_price[ord_vars] <- lapply(house_price[ord_vars], ordinal_to_numeric)
house_price <- na.omit(house_price)

cat_vars <- names(house_price)[sapply(house_price, function(x) is.factor(x) || is.character(x))]
for (v in cat_vars) {
  cat("\n-----\n")
  cat("Variable:", v, "\n")
  print(table(house_price[[v]], useNA = "ifany"))
}

```

```

##
## -----
## Variable: MSSubClass
##
## 120 160 180 190 20 30 40 45 50 60 70 75 80 85 90
## 67 53 6 19 425 47 3 9 116 230 52 13 37 14 35
##
## -----
## Variable: MSZoning
##
## C (all) FV RH RL RM
## 8 57 10 876 175
##
## -----
## Variable: Street
##
## Grvl Pave
## 4 1122
##
## -----
## Variable: Alley

```

```

##
##      Grvl NoAlley      Pave
##      41      1048      37
##
## -----
## Variable: LandContour
##
##      Bnk  HLS  Low  Lvl
##      47   44   15 1020
##
## -----
## Variable: Utilities
##
## AllPub
##      1126
##
## -----
## Variable: LotConfig
##
##      Corner CulDSac      FR2      FR3  Inside
##      191      44      30      4      857
##
## -----
## Variable: Neighborhood
##
## Blmngtn Blueste  BrDale BrkSide ClearCr CollgCr Crawfor Edwards Gilbert  IDOTRR
##      14      2      15      42      12      124      41      71      49      27
## MeadowV Mitchel  NAmes NoRidge NPKvill NridgHt  NWAmes OldTown  Sawyer SawyerW
##      10      32      181      33      7      76      45      97      46      49
## Somerst StoneBr  SWISU  Timber Veenker
##      78      20      19      29      7
##
## -----
## Variable: Condition1
##
## Artery  Feedr  Norm  PosA  PosN  RRAe  RRAn  RRNe  RRNn
##      43      56      976      4      9      9      24      1      4
##
## -----
## Variable: Condition2
##
## Artery  Feedr  Norm  PosA  PosN  RRNn
##      2      5      1114      1      2      2
##
## -----
## Variable: BldgType
##
##      1Fam 2fmCon Duplex  Twnhs TwnhsE
##      945      20      35      35      91
##
## -----
## Variable: HouseStyle
##
## 1.5Fin 1.5Unf 1Story 2.5Fin 2.5Unf 2Story SFoyer  SLvl

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```

##      121      10      563      6      9      350      24      43
##
## -----
## Variable: RoofStyle
##
##      Flat      Gable Gambrel      Hip Mansard
##          6      870      10      234      6
##
## -----
## Variable: RoofMatl
##
## ClyTile CompShg Membran      Roll Tar&Grv WdShake WdShngl
##          1      1109      1      1      6      2      6
##
## -----
## Variable: Exterior1st
##
## AsbShng BrkComm BrkFace CBlock CemntBd HdBoard ImStucc MetalSd Plywood Stone
##          14          2      40          1      47      155          1      175      68          1
## Stucco VinylSd Wd Sdng WdShng
##          19      430      151      22
##
## -----
## Variable: Exterior2nd
##
## AsbShng AsphShn Brk Cmn BrkFace CBlock CmentBd HdBoard ImStucc MetalSd Other
##          15          2          7      21          1      47      142          9      172          1
## Plywood Stone Stucco VinylSd Wd Sdng Wd Shng
##          87          4      20      421      148      29
##
## -----
## Variable: MasVnrType
##
##          BrkCmn      BrkFace NoMasVnrType      None      Stone
##          9          333          6          659          119
##
## -----
## Variable: Foundation
##
## BrkTil CBlock PConc Slab Stone Wood
##          122      453      528      15      6      2
##
## -----
## Variable: Heating
##
## GasA GasW Grav OthW Wall
##          1103      17      2      1      3
##
## -----
## Variable: CentralAir
##
##          N      Y
##          63 1063
##

```

```

## -----
## Variable: Electrical
##
## FuseA FuseF FuseP   Mix SBrkr
##   70   20   2     1  1033
##
## -----
## Variable: GarageType
##
## 2Types  Attchd Basment BuiltIn CarPort  Detchd
##     5    694    15     64     8    340
##
## -----
## Variable: Fence
##
##   GdPrv   GdWo   MnPrv   MnWw NoFence
##     47     39    119     10    911
##
## -----
## Variable: MiscFeature
##
## NoMiscFeature      Othr      Shed      TenC
##         1090         1         34         1
##
## -----
## Variable: SaleType
##
##   COD   Con ConLD ConLI ConLw   CWD   New   Oth   WD
##   32    2    6    3    4     4   119    1   955
##
## -----
## Variable: SaleCondition
##
## Abnorml AdjLand  Alloca  Family  Normal  Partial
##     71      1      9     18     905     122
##
## -----
## Variable: HasBsmt
##
##   0    1
##  24 1102
##
## -----
## Variable: HasGarage
##
##   0    1
##   0 1126
##
## -----
## Variable: HasFireplace
##
##   0    1
## 533 593
##

```

```

## -----
## Variable: HasPool
##
##    0    1
## 1120   6
##
## -----
## Variable: HasMasVnr
##
##    0    1
## 664 462

# month as time seires
house_price$month_sin <- sin(2 * pi * house_price$MoSold / 12)
house_price$month_cos <- cos(2 * pi * house_price$MoSold / 12)
#
house_price$YearSinceRemodel <- house_price$YrSold - house_price$YearRemodAdd

house_price <- house_price |>
  select(-Id, -MoSold, -YrSold, -YearRemodAdd,
         # only one level
         -HasGarage, -Utilities,
         # too sparse
         -Street, -Condition2, -RoofMatl, -Heating, -HasPool, -MiscFeature)

# merge some level
house_price$Condition1 <- dplyr::case_when(
  house_price$Condition1 == "Norm" ~ "Norm",
  house_price$Condition1 %in% c("Artery", "Feedr") ~ "Road",
  house_price$Condition1 %in% c("RRAn", "RRAe", "RRNe", "RRNn") ~ "Railroad",
  house_price$Condition1 %in% c("PosA", "PosN") ~ "PositiveFeature",
  TRUE ~ "Other"
)
house_price$Condition1 <- factor(house_price$Condition1)
house_price$RoofStyle <- dplyr::case_when(
  house_price$RoofStyle %in% c("Gable", "Hip") ~ house_price$RoofStyle,
  TRUE ~ "Other"
)
house_price$RoofStyle <- factor(house_price$RoofStyle)
house_price$Exterior1st <- dplyr::case_when(
  house_price$Exterior1st %in% c("VinylSd", "MetalSd", "HdBoard", "Wd Sdng", "Plywood", "CemntBd") ~ house_price$Exterior1st,
  house_price$Exterior1st %in% c("BrkFace", "Stone", "Stucco", "WdShng", "AsbShng") ~ house_price$Exterior1st,
  TRUE ~ "Other"
)
house_price$Exterior1st <- factor(house_price$Exterior1st)
house_price$Exterior2nd <- dplyr::case_when(
  house_price$Exterior2nd %in% c("VinylSd", "MetalSd", "HdBoard", "Wd Sdng", "Plywood", "CmentBd", "Wd Sdng") ~ house_price$Exterior2nd,
  house_price$Exterior2nd %in% c("BrkFace", "Stucco", "AsbShng") ~ house_price$Exterior2nd,
  TRUE ~ "Other"
)
house_price$Exterior2nd <- factor(house_price$Exterior2nd)
house_price$Electrical <- dplyr::case_when(
  house_price$Electrical == "SBrkr" ~ "SBrkr",
  house_price$Electrical %in% c("FuseA", "FuseF") ~ "Fuse",

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```

house_price$Electrical %in% c("FuseP", "Mix") ~ "PoorOrMixed",
TRUE ~ "Other"
)
house_price$Electrical <- factor(house_price$Electrical)
house_price$GarageType <- dplyr::case_when(
  house_price$GarageType == "Attchd" ~ "Attached",
  house_price$GarageType == "Detchd" ~ "Detached",
  house_price$GarageType == "BuiltIn" ~ "BuiltIn",
  house_price$GarageType %in% c("Basment", "CarPort", "2Types") ~ "OtherGarage",
  TRUE ~ "OtherGarage"
)
house_price$GarageType <- factor(house_price$GarageType)
house_price$SaleType <- dplyr::case_when(
  house_price$SaleType == "WD" ~ "WarrantyDeed",
  house_price$SaleType == "New" ~ "NewHome",
  house_price$SaleType == "COD" ~ "CourtOrEstate",
  house_price$SaleType %in% c("CWD", "Con", "ConLD", "ConLI", "ConLw", "Oth") ~ "OtherSaleType",
  TRUE ~ "OtherSaleType"
)
house_price$SaleType <- factor(house_price$SaleType)
house_price$SaleCondition <- dplyr::case_when(
  house_price$SaleCondition == "Normal" ~ "Normal",
  house_price$SaleCondition == "Abnorml" ~ "Abnormal",
  house_price$SaleCondition == "Partial" ~ "Partial",
  house_price$SaleCondition %in% c("AdjLand", "Alloca", "Family") ~ "OtherCondition",
  TRUE ~ "OtherCondition"
)
house_price$SaleCondition <- factor(house_price$SaleCondition)

cat_vars <- names(house_price)[sapply(house_price, function(x) is.factor(x) || is.character(x))]
for (v in cat_vars) {
  cat("\n-----\n")
  cat("Variable:", v, "\n")
  print(table(house_price[[v]], useNA = "ifany"))
}

```

```

##
## -----
## Variable: MSSubClass
##
## 120 160 180 190 20 30 40 45 50 60 70 75 80 85 90
## 67 53 6 19 425 47 3 9 116 230 52 13 37 14 35
##
## -----
## Variable: MSZoning
##
## C (all) FV RH RL RM
## 8 57 10 876 175
##
## -----
## Variable: Alley
##
## Grv1 NoAlley Pave
## 41 1048 37

```



```

##
## -----
## Variable: LandContour
##
## Bnk  HLS  Low  Lvl
##  47   44   15 1020
##
## -----
## Variable: LotConfig
##
## Corner CulDSac  FR2  FR3  Inside
##   191    44    30    4    857
##
## -----
## Variable: Neighborhood
##
## Blmngtn Blueste  BrDale BrkSide ClearCr CollgCr Crawfor Edwards Gilbert  IDOTRR
##    14      2    15    42    12    124    41    71    49    27
## MeadowV Mitchel  NAmes NoRidge NPKvill NridgHt  NWAmes OldTown  Sawyer SawyerW
##    10     32   181    33     7     76    45    97    46    49
## Somerst StoneBr  SWISU  Timber Veenker
##    78     20    19    29     7
##
## -----
## Variable: Condition1
##
##          Norm PositiveFeature          Railroad          Road
##          976              13              38              99
##
## -----
## Variable: BldgType
##
## 1Fam 2fmCon Duplex  Twnhs TwnhsE
##  945   20   35   35   91
##
## -----
## Variable: HouseStyle
##
## 1.5Fin 1.5Unf 1Story 2.5Fin 2.5Unf 2Story SFoyer  SLvl
##   121    10   563     6     9   350    24   43
##
## -----
## Variable: RoofStyle
##
## Gable  Hip Other
##   870   234   22
##
## -----
## Variable: Exterior1st
##
## AsbShng BrkFace CemntBd HdBoard MetalSd  Other Plywood  Stone  Stucco VinylSd
##    14    40    47    155    175     4    68     1    19   430
## Wd Sdng WdShing
##   151     22

```

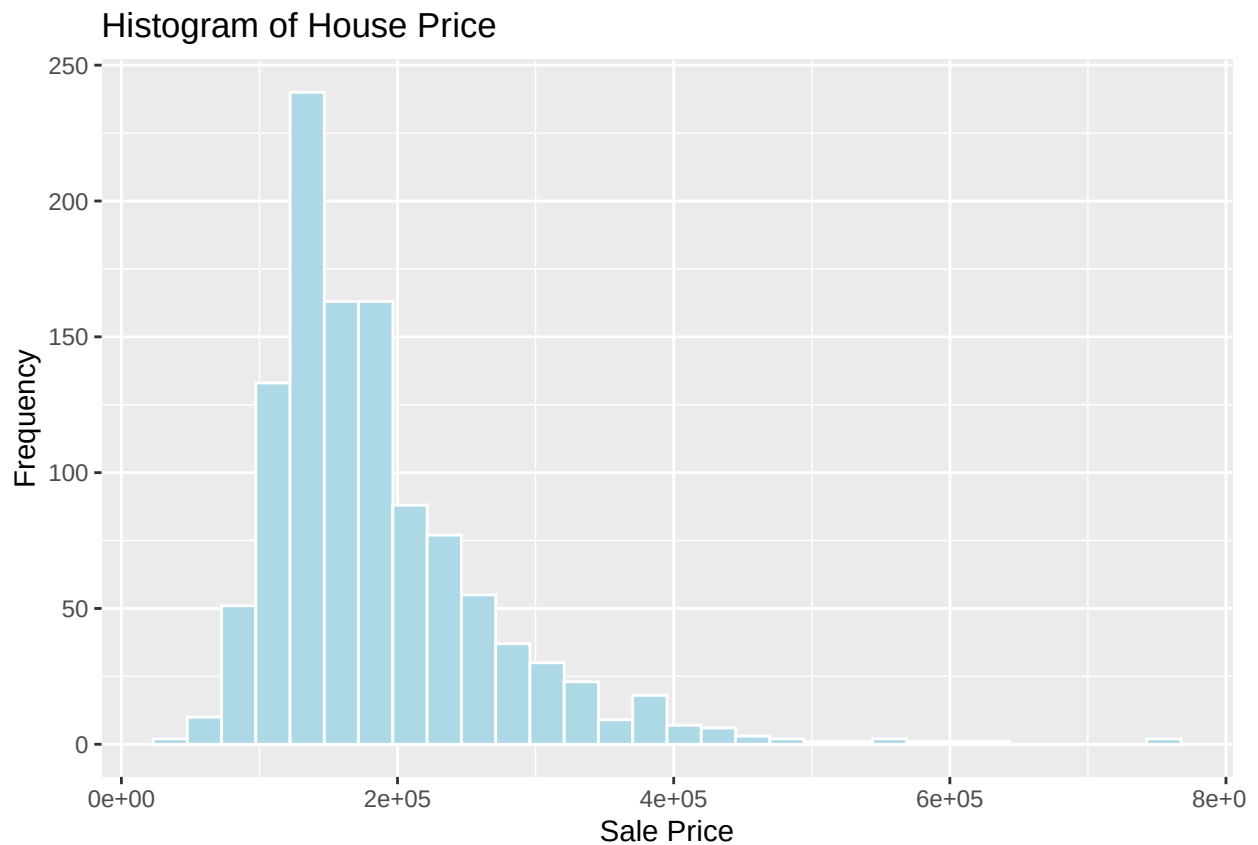
```

##
## -----
## Variable: Exterior2nd
##
## AsbShng BrkFace CmentBd HdBoard MetalSd Other Plywood Stucco VinylSd Wd Sdng
##      15      21      47      142      172      24      87      20      421      148
## Wd Shng
##      29
##
## -----
## Variable: MasVnrType
##
##      BrkCmn      BrkFace NoMasVnrType      None      Stone
##          9          333          6          659          119
##
## -----
## Variable: Foundation
##
## BrkTil CBlock PConc Slab Stone Wood
##      122      453      528      15      6      2
##
## -----
## Variable: CentralAir
##
##      N      Y
##      63 1063
##
## -----
## Variable: Electrical
##
##      Fuse PoorOrMixed      SBrkr
##          90          3          1033
##
## -----
## Variable: GarageType
##
##      Attached      BuiltIn      Detached OtherGarage
##          694          64          340          28
##
## -----
## Variable: Fence
##
##      GdPrv      GdWo      MnPrv      MnWw NoFence
##          47          39          119          10          911
##
## -----
## Variable: SaleType
##
## CourtOrEstate      NewHome OtherSaleType WarrantyDeed
##          32          119          20          955
##
## -----
## Variable: SaleCondition
##

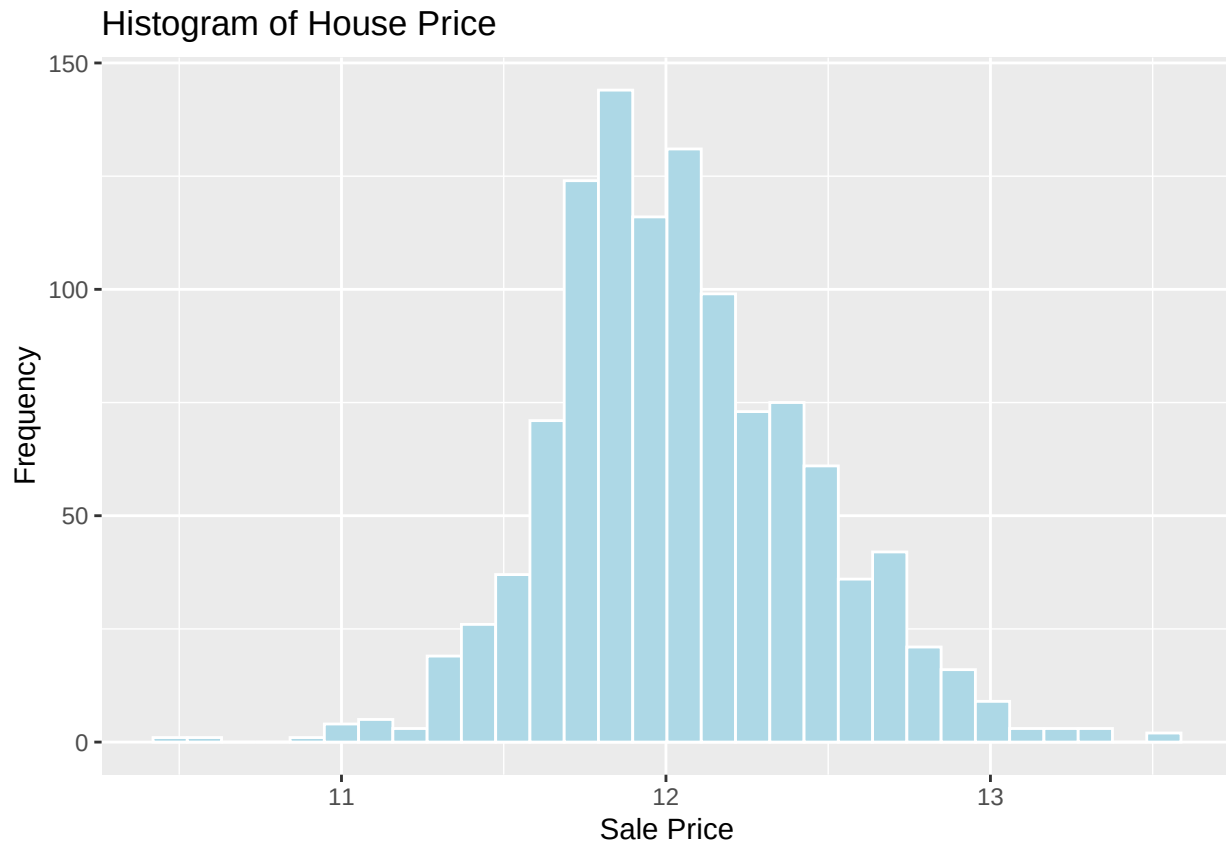
```

```
##      Abnormal      Normal OtherCondition      Partial
##           71          905           28          122
##
## -----
## Variable: HasBsmt
##
##    0    1
##  24 1102
##
## -----
## Variable: HasFireplace
##
##    0    1
## 533 593
##
## -----
## Variable: HasMasVnr
##
##    0    1
## 664 462
```

```
# check distribution of Y
ggplot(house_price, aes(x = SalePrice)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white") +
  labs(title = "Histogram of House Price",
       x = "Sale Price",
       y = "Frequency")
```



```
# Right skew -> log(SalePrice)
house_price$LogSalePrice = log(house_price$SalePrice)
ggplot(house_price, aes(x = LogSalePrice)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white") +
  labs(title = "Histogram of House Price",
       x = "Sale Price",
       y = "Frequency")
```



```
house_price <- house_price |> select(-SalePrice)
```

```
numeric_vars <- house_price[, sapply(house_price, is.numeric)]
cor_matrix <- cor(numeric_vars, use = "pairwise.complete.obs")
cor_with_log_price_abs <- abs(cor_matrix[, "LogSalePrice"])
cor_with_log_price_abs <- sort(
  cor_with_log_price_abs[names(cor_with_log_price_abs) != "LogSalePrice"],
  decreasing = TRUE
)
cor_with_log_price_abs
```

##	OverallQual	ExterQual	GrLivArea	KitchenQual
##	0.829026568	0.709547984	0.695477575	0.686360986
##	GarageCars	BsmtQual	GarageArea	TotalBsmtSF
##	0.665407947	0.647958389	0.627927567	0.618239836
##	FullBath	X1stFlrSF	YearSinceRemodel	YearBuilt
##	0.609465503	0.604950177	0.595291666	0.594798212
##	GarageFinish	GarageYrBlt	TotRmsAbvGrd	FireplaceQu

```
##      0.585711121      0.564066814      0.542626220      0.539809103
##      HeatingQC      Fireplaces      MasVnrArea      BsmtExposure
##      0.506222945      0.480893101      0.440431885      0.383174268
##      OpenPorchSF      BsmtFinSF1      LotFrontage      WoodDeckSF
##      0.373178823      0.372763649      0.354562115      0.346075611
##      BsmtFinType1      LotShape      X2ndFlrSF      HalfBath
##      0.335161548      0.332308393      0.298185483      0.298130317
##      LotArea      BsmtCond      BsmtFullBath      PavedDrive
##      0.281460581      0.246754055      0.243844335      0.232468012
##      BsmtUnfSF      BedroomAbvGr      GarageQual      EnclosedPorch
##      0.229931247      0.196913859      0.188022328      0.177411889
##      GarageCond      KitchenAbvGr      Functional      ScreenPorch
##      0.173775411      0.157999818      0.120560126      0.118121121
##      OverallCond      PoolQC      month_sin      PoolArea
##      0.093206171      0.077044913      0.067270916      0.065771781
##      month_cos      X3SsnPorch      MiscVal      BsmtHalfBath
##      0.046253492      0.037988471      0.035300817      0.032790298
##      LandSlope      BsmtFinSF2      BsmtFinType2      LowQualFinSF
##      0.026497431      0.019182272      0.018075556      0.013911951
##      ExterCond
##      0.003460266
```

```
drop_vars <- names(cor_with_log_price_abs[cor_with_log_price_abs < 0.3])
house_price <- house_price |> select(-all_of(drop_vars))
```

```
# drop high cor features
numeric_vars2 <- house_price[, sapply(house_price, is.numeric)]
numeric_vars2 <- numeric_vars2[, colnames(numeric_vars2) != "LogSalePrice"]
selected_cor_matrix <- cor(numeric_vars2, use = "pairwise.complete.obs")
high_cor_pairs <- which(
  abs(selected_cor_matrix) > 0.7 & upper.tri(selected_cor_matrix),
  arr.ind = TRUE
)
high_cor_df <- data.frame(
  var1 = rownames(selected_cor_matrix)[high_cor_pairs[, 1]],
  var2 = colnames(selected_cor_matrix)[high_cor_pairs[, 2]],
  cor = selected_cor_matrix[high_cor_pairs]
)
high_cor_df
```

```
##      var1      var2      cor
## 1 OverallQual      ExterQual 0.7498563
## 2 TotalBsmtSF      X1stFlrSF 0.8366715
## 3      ExterQual      KitchenQual 0.7324666
## 4      GrLivArea      TotRmsAbvGrd 0.8239077
## 5      Fireplaces      FireplaceQu 0.8677742
## 6      YearBuilt      GarageYrBlt 0.8243722
## 7      GarageCars      GarageArea 0.8400285
```

```
drop_vars_highcor <- c()

for (i in 1:nrow(high_cor_df)) {
  v1 <- high_cor_df$var1[i]
  v2 <- high_cor_df$var2[i]
```

```

c1 <- cor_with_log_price_abs[v1]
c2 <- cor_with_log_price_abs[v2]

if (c1 < c2) {
  drop_vars_highcor <- c(drop_vars_highcor, v1)
} else {
  drop_vars_highcor <- c(drop_vars_highcor, v2)
}
}

drop_vars_highcor <- unique(drop_vars_highcor)
house_price <- house_price |> select(-all_of(drop_vars_highcor))

# check categorical features using ANOVA
# convert character to factor
house_price[] <- lapply(house_price, function(x) {
  if (is.character(x)) as.factor(x) else x
})
# select factor variables only
cat_features <- house_price[, sapply(house_price, is.factor)]

cat_features <- cat_features[, colnames(cat_features) != "LogSalePrice", drop = FALSE]
cat_pvalues_list <- lapply(names(cat_features), function(var) {
  fit <- aov(as.formula(paste("LogSalePrice ~", var)), data = house_price)
  p_val <- summary(fit)[[1]][["Pr(>F)"]][1]

  level_counts <- table(house_price[[var]])

  data.frame(
    variable = var,
    p_value = p_val,
    n_levels = nlevels(cat_features[[var]]),
    min_count = min(level_counts),
    stringsAsFactors = FALSE
  )
})
cat_pvalues <- do.call(rbind, cat_pvalues_list)
cat_pvalues <- cat_pvalues[order(cat_pvalues$p_value), ]
cat_pvalues

```

##	variable	p_value	n_levels	min_count
## 6	Neighborhood	2.176243e-189	25	2
## 14	Foundation	3.579833e-96	6	2
## 1	MSSubClass	5.482285e-83	15	3
## 17	GarageType	7.951334e-79	4	28
## 22	HasFireplace	1.564175e-72	2	533
## 13	MasVnrType	2.839508e-56	5	6
## 11	Exterior1st	6.552983e-53	12	1
## 12	Exterior2nd	1.637554e-50	11	15
## 2	MSZoning	6.295085e-46	5	8
## 23	HasMasVnr	6.464496e-43	2	462
## 20	SaleCondition	1.018836e-39	4	28
## 19	SaleType	8.419164e-38	4	20
## 15	CentralAir	1.206114e-25	2	63

```
## 16    Electrical  2.542585e-24      3      3
## 9      HouseStyle 1.238701e-22      8      6
## 10     RoofStyle  8.083170e-13      3     22
## 18      Fence    1.146946e-10      5     10
## 8      BldgType   1.000710e-09      5     20
## 7     Condition1 8.022774e-09      4     13
## 5      LotConfig  2.263226e-08      5      4
## 21     HasBsmt    2.988616e-08      2     24
## 3      Alley     6.203825e-08      3     37
## 4    LandContour  5.782590e-07      4     15
```

```
#full anova model
```

```
full_model <- lm(LogSalePrice ~ ., data = house_price)
alias(full_model)
```

```
## Model :
```

```
## LogSalePrice ~ MSSubClass + MSZoning + LotFrontage + Alley +
##   LotShape + LandContour + LotConfig + Neighborhood + Condition1 +
##   BldgType + HouseStyle + OverallQual + YearBuilt + RoofStyle +
##   Exterior1st + Exterior2nd + MasVnrType + MasVnrArea + Foundation +
##   BsmtQual + BsmtExposure + BsmtFinType1 + BsmtFinSF1 + TotalBsmtSF +
##   HeatingQC + CentralAir + Electrical + GrLivArea + FullBath +
##   FireplaceQu + GarageType + GarageFinish + GarageCars + WoodDeckSF +
##   OpenPorchSF + Fence + SaleType + SaleCondition + HasBsmt +
##   HasFireplace + HasMasVnr + YearSinceRemodel
##
```

```
## Complete :
```

```
##           (Intercept) MSSubClass160 MSSubClass180 MSSubClass190
## BldgTypeDuplex      0           0           0           0
## BldgTypeTwnhsE      1           0           0          -1
##           MSSubClass20 MSSubClass30 MSSubClass40 MSSubClass45 MSSubClass50
## BldgTypeDuplex      0           0           0           0           0
## BldgTypeTwnhsE     -1          -1          -1          -1          -1
##           MSSubClass60 MSSubClass70 MSSubClass75 MSSubClass80 MSSubClass85
## BldgTypeDuplex      0           0           0           0           0
## BldgTypeTwnhsE     -1          -1          -1          -1          -1
##           MSSubClass90 MSZoningFV MSZoningRH MSZoningRL MSZoningRM
## BldgTypeDuplex      1           0           0           0           0
## BldgTypeTwnhsE     -1           0           0           0           0
##           LotFrontage AlleyNoAlley AlleyPave LotShape LandContourHLS
## BldgTypeDuplex      0           0           0           0           0
## BldgTypeTwnhsE      0           0           0           0           0
##           LandContourLow LandContourLvl LotConfigCulDSac LotConfigFR2
## BldgTypeDuplex      0           0           0           0
## BldgTypeTwnhsE      0           0           0           0
##           LotConfigFR3 LotConfigInside NeighborhoodBlueste
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodBrDale NeighborhoodBrkSide NeighborhoodClearCr
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodCollgCr NeighborhoodCrawfor NeighborhoodEdwards
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodGilbert NeighborhoodIDOTRR NeighborhoodMeadowV
```

##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		NeighborhoodMitchel	NeighborhoodNames	NeighborhoodNoRidge
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		NeighborhoodNPkVill	NeighborhoodNridgHt	NeighborhoodNWAmes
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		NeighborhoodOldTown	NeighborhoodSawyer	NeighborhoodSawyerW
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		NeighborhoodSomerst	NeighborhoodStoneBr	NeighborhoodSWISU
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		NeighborhoodTimber	NeighborhoodVeenker	Condition1PositiveFeature
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Condition1Railroad	Condition1Road	BldgType2fmCon BldgTypeTwnhs
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	-1
##		HouseStyle1.5Unf	HouseStyle1Story	HouseStyle2.5Fin
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		HouseStyle2.5Unf	HouseStyle2Story	HouseStyleSFoyer
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		HouseStyleSLvl	OverallQual	YearBuilt RoofStyleHip RoofStyleOther
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stBrkFace	Exterior1stCemntBd	Exterior1stHdBoard
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stMetalSd	Exterior1stOther	Exterior1stPlywood
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stStone	Exterior1stStucco	Exterior1stVinylSd
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stWd Sdng	Exterior1stWdShing	Exterior2ndBrkFace
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior2ndCmentBd	Exterior2ndHdBoard	Exterior2ndMetalSd
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior2ndOther	Exterior2ndPlywood	Exterior2ndStucco
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior2ndVinylSd	Exterior2ndWd Sdng	Exterior2ndWd Shng
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		MasVnrTypeBrkFace	MasVnrTypeNoMasVnrType	MasVnrTypeNone
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		MasVnrTypeStone	MasVnrArea	FoundationCBlock FoundationPConc


```

## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## FoundationSlab FoundationStone FoundationWood BsmtQual
## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## BsmtExposure BsmtFinType1 BsmtFinSF1 TotalBsmtSF HeatingQC
## BldgTypeDuplex 0 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0 0
## CentralAirY ElectricalPoorOrMixed ElectricalSBrkr GrLivArea
## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## FullBath FireplaceQu GarageTypeBuiltIn GarageTypeDetached
## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## GarageTypeOtherGarage GarageFinish GarageCars WoodDeckSF
## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## OpenPorchSF FenceGdWo FenceMnPrv FenceMnWw FenceNoFence
## BldgTypeDuplex 0 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0 0
## SaleTypeNewHome SaleTypeOtherSaleType SaleTypeWarrantyDeed
## BldgTypeDuplex 0 0 0
## BldgTypeTwnhsE 0 0 0
## SaleConditionNormal SaleConditionOtherCondition
## BldgTypeDuplex 0 0
## BldgTypeTwnhsE 0 0
## SaleConditionPartial HasBsmt1 HasFireplace1 HasMasVnr1
## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## YearSinceRemodel
## BldgTypeDuplex 0
## BldgTypeTwnhsE 0

```

```

house_price <- house_price|> select(-BldgType)
reduced_model <- lm(LogSalePrice ~ ., data = house_price)
anova(reduced_model)

```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: LogSalePrice
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
## MSSubClass	14	56.743	4.0531	250.7686	< 2.2e-16 ***
## MSZoning	4	8.791	2.1977	135.9718	< 2.2e-16 ***
## LotFrontage	1	8.251	8.2508	510.4846	< 2.2e-16 ***
## Alley	2	0.199	0.0995	6.1545	0.0022057 **
## LotShape	1	3.932	3.9321	243.2852	< 2.2e-16 ***
## LandContour	3	1.410	0.4701	29.0829	< 2.2e-16 ***
## LotConfig	4	2.403	0.6007	37.1665	< 2.2e-16 ***
## Neighborhood	24	42.587	1.7745	109.7881	< 2.2e-16 ***
## Condition1	3	0.326	0.1086	6.7221	0.0001718 ***
## HouseStyle	7	0.684	0.0978	6.0491	6.291e-07 ***
## OverallQual	1	19.077	19.0769	1180.3136	< 2.2e-16 ***
## YearBuilt	1	0.692	0.6917	42.7934	9.721e-11 ***
## RoofStyle	2	0.275	0.1373	8.4927	0.0002202 ***
## Exterior1st	11	1.316	0.1197	7.4037	2.717e-12 ***

```
## Exterior2nd      10  0.335  0.0335    2.0722 0.0241342 *
## MasVnrType       4  0.373  0.0932    5.7672 0.0001370 ***
## MasVnrArea       1  0.259  0.2588   16.0134 6.757e-05 ***
## Foundation       5  0.536  0.1071    6.6278 4.421e-06 ***
## BsmtQual         1  1.013  1.0133   62.6967 6.420e-15 ***
## BsmtExposure     1  0.622  0.6220   38.4864 8.079e-10 ***
## BsmtFinType1     1  1.190  1.1899   73.6182 < 2.2e-16 ***
## BsmtFinSF1       1  0.300  0.2996   18.5376 1.831e-05 ***
## TotalBsmtSF      1  0.952  0.9519   58.8975 3.957e-14 ***
## HeatingQC        1  0.643  0.6426   39.7569 4.320e-10 ***
## CentralAir       1  0.600  0.5998   37.1112 1.593e-09 ***
## Electrical       2  0.306  0.1529    9.4614 8.503e-05 ***
## GrLivArea        1  4.748  4.7485  293.7942 < 2.2e-16 ***
## FullBath         1  0.122  0.1217    7.5276 0.0061856 **
## FireplaceQu      1  0.368  0.3676   22.7423 2.129e-06 ***
## GarageType       3  0.039  0.0128    0.7942 0.4971816
## GarageFinish     1  0.118  0.1182    7.3132 0.0069619 **
## GarageCars       1  1.202  1.2016   74.3434 < 2.2e-16 ***
## WoodDeckSF       1  0.077  0.0768    4.7520 0.0294976 *
## OpenPorchSF      1  0.123  0.1225    7.5819 0.0060031 **
## Fence           4  0.080  0.0201    1.2417 0.2914920
## SaleType         3  0.135  0.0449    2.7806 0.0400025 *
## SaleCondition    3  0.296  0.0987    6.1086 0.0004061 ***
## HasBsmt          1  0.005  0.0052    0.3199 0.5717881
## HasFireplace     1  0.011  0.0111    0.6891 0.4066665
## HasMasVnr        1  0.014  0.0141    0.8694 0.3513405
## YearSinceRemodel  1  0.286  0.2864   17.7181 2.794e-05 ***
## Residuals       994 16.066  0.0162
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
house_price <- house_price |>
  select(
    -GarageType,
    -Fence,
    -HasBsmt,
    -HasFireplace,
    -HasMasVnr,
    # *
    -Exterior2nd,
    -WoodDeckSF,
    -SaleType,
    # **
    -Alley,
    -FullBath,
    -GarageFinish,
    -OpenPorchSF
  )
full_model <- lm(LogSalePrice ~ ., data = house_price)
vif(full_model)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## MSSubClass    3.803055e+07 14      1.865173
## MSZoning      3.864501e+01  4      1.579017
## LotFrontage   2.465802e+00  1      1.570287
```

```
## LotShape      1.590447e+00  1      1.261129
## LandContour   2.205382e+00  3      1.140899
## LotConfig     1.875354e+00  4      1.081771
## Neighborhood  3.136809e+04 24      1.240729
## Condition1    1.862151e+00  3      1.109181
## HouseStyle    9.293039e+05  7      2.668683
## OverallQual   4.710619e+00  1      2.170396
## YearBuilt     1.391338e+01  1      3.730064
## RoofStyle     1.687911e+00  2      1.139823
## Exterior1st   2.716171e+01 11      1.161930
## MasVnrType     4.511194e+00  4      1.207220
## MasVnrArea    2.671804e+00  1      1.634565
## Foundation    2.533155e+01  5      1.381549
## BsmtQual      4.313099e+00  1      2.076800
## BsmtExposure   1.921678e+00  1      1.386246
## BsmtFinType1   2.691768e+00  1      1.640661
## BsmtFinSF1     3.128073e+00  1      1.768636
## TotalBsmtSF    5.800174e+00  1      2.408355
## HeatingQC     2.154476e+00  1      1.467813
## CentralAir    1.706733e+00  1      1.306420
## Electrical    1.737151e+00  2      1.148046
## GrLivArea     6.813257e+00  1      2.610222
## FireplaceQu   1.927492e+00  1      1.388341
## GarageCars    2.445604e+00  1      1.563843
## SaleCondition  2.054043e+00  3      1.127461
## YearSinceRemodel 2.656234e+00  1      1.629796
```

```
str(house_price)
```

```
## 'data.frame':    1126 obs. of  30 variables:
## $ MSSubClass      : Factor w/ 15 levels "120","160","180",...: 10 5 10 11 10 9 5 9 4 5 ...
## $ MSZoning        : Factor w/ 5 levels "C (all)","FV",...: 4 4 4 4 4 4 4 5 4 4 ...
## $ LotFrontage     : int   65 80 68 60 84 85 75 51 50 70 ...
## $ LotShape        : num   3 3 2 2 2 2 3 3 3 3 ...
## $ LandContour     : Factor w/ 4 levels "Bnk","HLS","Low",...: 4 4 4 4 4 4 4 4 4 4 ...
## $ LotConfig       : Factor w/ 5 levels "Corner","CulDSac",...: 5 3 5 1 3 5 5 5 1 5 ...
## $ Neighborhood    : Factor w/ 25 levels "Blmngtn","Blueste",...: 6 25 6 7 14 12 21 18 4 19 ...
## $ Condition1      : Factor w/ 4 levels "Norm","PositiveFeature",...: 1 4 1 1 1 1 1 4 4 1 ...
## $ HouseStyle       : Factor w/ 8 levels "1.5Fin","1.5Unf",...: 6 3 6 6 6 1 3 1 2 3 ...
## $ OverallQual      : int   7 6 7 7 8 5 8 7 5 5 ...
## $ YearBuilt        : int  2003 1976 2001 1915 2000 1993 2004 1931 1939 1965 ...
## $ RoofStyle        : Factor w/ 3 levels "Gable","Hip",...: 1 1 1 1 1 1 1 1 1 2 ...
## $ Exterior1st      : Factor w/ 12 levels "AsbShng","BrkFace",...: 10 5 10 11 10 10 10 2 5 4 ...
## $ MasVnrType       : Factor w/ 5 levels "BrkCmn","BrkFace",...: 2 4 2 4 2 4 5 4 4 4 ...
## $ MasVnrArea       : num   196 0 162 0 350 0 186 0 0 0 ...
## $ Foundation       : Factor w/ 6 levels "BrkTil","CBlock",...: 3 2 3 1 3 6 3 1 1 2 ...
## $ BsmtQual         : num   4 4 4 3 4 4 5 3 3 3 ...
## $ BsmtExposure     : num   1 4 2 1 3 1 3 1 1 1 ...
## $ BsmtFinType1     : num   6 5 6 5 6 6 6 1 6 3 ...
## $ BsmtFinSF1       : int   706 978 486 216 655 732 1369 0 851 906 ...
## $ TotalBsmtSF      : int   856 1262 920 756 1145 796 1686 952 991 1040 ...
## $ HeatingQC        : num   4 4 4 3 4 4 4 3 4 4 ...
## $ CentralAir       : Factor w/ 2 levels "N","Y": 2 2 2 2 2 2 2 2 2 2 ...
## $ Electrical       : Factor w/ 3 levels "Fuse","PoorOrMixed",...: 3 3 3 3 3 3 3 1 3 3 ...
## $ GrLivArea        : int  1710 1262 1786 1717 2198 1362 1694 1774 1077 1040 ...
```

```
## $ FireplaceQu      : num  0 3 3 4 3 0 4 3 3 0 ...
## $ GarageCars       : int   2 2 2 3 3 2 2 2 1 1 ...
## $ SaleCondition    : Factor w/ 4 levels "Abnormal","Normal",...: 2 2 2 1 2 2 2 1 2 2 ...
## $ YearSinceRemodel: int    5 31 6 36 8 14 2 58 58 43 ...
## $ LogSalePrice     : num   12.2 12.1 12.3 11.8 12.4 ...
## - attr(*, "na.action")= 'omit' Named int [1:334] 8 13 15 17 25 32 40 43 44 49 ...
##   ..- attr(*, "names")= chr [1:334] "8" "13" "15" "17" ...
```

Model

native model(only consider some numeric variable)

$$y_i = \log(\text{SalePrice}_i)$$

$$y_i \sim \mathcal{N}(\mu_i, \sigma^2),$$

$$\mu_i = \beta_0 + \beta_1 \text{OverallQual}_i + \beta_2 \text{GrLivArea}_i + \beta_3 \text{GarageCars}_i + \beta_4 \text{TotalBsmtSF}_i + \beta_5 \text{YearBuilt}_i + \beta_6 \text{YearSinceRemodel}_i.$$

complex model